Dominik Stiftinger-Lang

Senior Backend & Financial Systems Engineer
Go, Digital Assets, Quantitative Finance & Applied AI

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Profile summary

Senior engineer, mathematician and founder with nine years in financial and digital-asset systems. Built distributed Go services, trading infrastructure, production ML, and AI-assisted delivery while leading products, projects, and companies.

Open to substantial engineering roles and engagements

Expertise

- Go & Distributed Systems
- Trading Systems & Risk
- Quantitative Finance
- Machine Learning & Research
- AI-Assisted Engineering
- Product & Company Leadership

Engagement

Available
1 September 2026

Capacity
30-40 hours per week

Work model
Remote, EU time zone

Languages

German	Native
English	Fluent

Selected Projects

2022 - present
ChainBrain FlexCo

Distributed Portfolio Management and Trading Platform

Co-Founder, CEO, Lead Backend Engineer and Architect

- Architected production Go services spanning portfolio construction, account state, execution, reconciliation, reporting, and client APIs.
- Used Temporal, PostgreSQL, gRPC, and OpenAPI for durable workflows and idempotent financial processing.
- Led product delivery and defined MiCA-aware custody, authorization, execution, analytics, and audit boundaries.

2024 - 2026
ChainBrain FlexCo

Multi-Venue Digital Asset Execution

Lead Engineer

- Built integrations for five centralized exchanges, including venue-specific order semantics and recovery behaviour.
- Developed best-execution algorithms, order-lifecycle state machines, a B-tree orderbook, reconciliation, exposure controls, and execution measurement.
- Operated spot and futures rebalancing with sub-1 bps median implementation shortfall and no exposure-limit breaches.

2020 - present
Nerox / Universities

Production Machine Learning and Quantitative Research

ML Engineer / Quantitative Researcher

- Built the Apex pipeline from real-time market data and feature engineering through TensorFlow/Keras training, backtesting, and staged live trading.
- Published spiking-neural-network research and developed actor-critic portfolio-optimization experiments under DeFi constraints.
- Used TensorFlow, Keras, PyTorch, Gym, Python, R, SQL, and production Go data services.

Technical Focus

- Go and Python
- Trading and Best Execution
- AI Agents and MCP
- Distributed Systems
- Quantitative Finance and ML
- Product Ownership and Scrum

Core Technologies

Languages & Quant Research
Go / Python / SQL
TensorFlow / Keras / PyTorch
Machine learning
Reinforcement learning

Trading & Digital Assets
Exchange APIs / WebSocket
HFT / Arbitrage / Trend / Grid
DeFi / Best execution
Order lifecycle / Reconciliation

Architecture & Delivery
Distributed systems / Temporal
PostgreSQL / gRPC / OpenAPI
Docker / GitLab CI
Product ownership / Scrum

Project History

ChainBrain FlexCo, Graz

2022 - present

Co-Founder, CEO, Lead Backend Engineer and Architect

- Co-founded ChainBrain within Nerox, operating under the ChainBrain brand from 2023 and as an independent company from March 2025.
- Established Scrum-based product and engineering delivery and initially facilitated it as Scrum Master. After the responsibilities were separated in 2023, focused on Product Ownership, stakeholder alignment, backlog priorities, sprint scope, and coordinated delivery.
- Wrote the project plan for an FFG-funded R&D project and managed delivery through final reporting.
- Architected and implemented a production digital-asset platform using distributed Go services, Temporal workflows, PostgreSQL, gRPC, OpenAPI, and idempotent financial processing.
- Designed service and data boundaries for portfolio construction, account state, execution, reconciliation, reporting, and client-facing APIs.
- Implemented Docker and GitLab CI delivery, Vault and Consul infrastructure, monitoring, incident response, and controlled secrets handling.
- Defined a MiCA-aware architecture with explicit custody, authorization, customer-controlled execution, hosted analytics, and audit boundaries.

Technologies: Go, Python, PostgreSQL, Temporal, gRPC, OpenAPI, REST, Hasura, Docker, GitLab CI, Vault, Consul

ChainBrain FlexCo

2024 - 2026

Selected Project - Multi-Venue Digital Asset Order Management and Execution

- Within ChainBrain, led REST and WebSocket integrations for Binance, Kraken, Bitfinex, Poloniex, and OKX, including venue-specific order semantics and recovery behaviour.
- Developed best-execution algorithms, order-lifecycle state machines, fill and balance reconciliation, exposure controls, a B-tree orderbook, and execution-quality measurement.
- Operated production spot and futures rebalancing with sub-1 bps median implementation shortfall and no exposure-limit breaches.

Technologies: Go, PostgreSQL, Temporal, REST, WebSocket, gRPC, Docker, best execution, implementation shortfall

ChainBrain FlexCo, Graz

2026 - present

Selected Initiative - AI-Assisted Engineering and Quality Control

- Designed multi-agent workflows with bounded roles for implementation, codebase research, test generation, security and financial-flow audits, and independent verification.
- Standardized Claude Code and OpenAI Codex through reusable skills, task specifications, structured handoffs, and persistent engineering knowledge.
- Integrated MCP-based tools for knowledge retrieval and operational workflows and designed evidence-based quality controls with traceable branch verdicts.

Technologies: Claude Code, OpenAI Codex, MCP, Go, Python, Bash, GitLab CI, Linux

:: Further Experience

Nerox GmbH, Graz

2019 - present

Co-Founder, Head of Quantitative Finance, later Chief Risk Officer

- Co-designed and built Apex (2020-2021), an autonomous digital-asset portfolio-management pipeline spanning real-time Go/WebSocket market-data collection, PostgreSQL storage, model deployment, and live trading.
- Implemented R/SQL/Python feature engineering and TensorFlow/Keras training with a ResNet-derived allocation model, portfolio-value objective, automated hyperparameter tuning, validation, backtesting, and staged capital rollout.
- Designed and operated HFT, arbitrage, trend-following, grid-trading, and DeFi strategies, with responsibility for risk controls, validation, deployment, and production performance.
- Owned quantitative product decisions and communicated model behaviour and risk to business stakeholders.
- Established and facilitated Scrum-based delivery across software and quantitative work, coordinating roadmap priorities, sprint delivery, and architecture decisions.

Technologies: Python, TensorFlow, Keras, R, PostgreSQL, SQL, Go, WebSocket, HFT, arbitrage, trend following, grid trading, DeFi, machine learning, reinforcement learning

Nerox GmbH, Graz

2017 - 2019

Co-Founder, Software Developer

- Implemented HFT, arbitrage, trend-following, and grid-trading strategies plus low-latency Go exchange adapters across Bybit, HitBTC, Bitstamp, Bittrex, KuCoin, Deribit, Huobi and other venues using REST and WebSocket market-data and order APIs.

Technologies: Go, REST, WebSocket, SQL, Linux, Bybit, HitBTC, Bitstamp, Bittrex, KuCoin, Deribit, Huobi, HFT, arbitrage, trend following, grid trading, low-latency systems

Graz University of Technology

2020 - 2021

Research Assistant - Spiking Neural Networks

- Developed and evaluated spiking-neural-network models in computational neuroscience; co-authored the resulting 2021 publication with Christoph Stoeckl and Wolfgang Maass.

Technologies: PyTorch, spiking neural networks, recurrent neural networks, PyBullet, evolutionary strategies, computational neuroscience, high-performance computing

:: Education

In progress	**University of Graz** PhD Mathematics Portfolio optimization under DeFi staking and lock-up constraints. Developed reinforcement-learning experiments from 2022, including a Gym environment and a TensorFlow/Keras actor-critic implementation.
2017 - 2019	**Graz University of Technology** MSc Mathematics Numerical methods for partial differential equations
2013 - 2017	**Graz University of Technology** BSc Mathematics, Technomathematics

:: Selected Publications

2025	**A Multi-Factor Investment Framework for Crypto Asset Management** Crypto Valley Association Research Journal. Highest reviewer rating among 12 selected papers.
2021	**Structure induces computational function in networks with diverse types of spiking neurons** Stoeckl, Lang, Maass. bioRxiv 10.1101/2021.05.18.444689.
2019	**A Study of Boundary Element Methods for Electrostatic Transmission Problems** MSc thesis, Graz University of Technology.